FORECASTING - PRACTICE PROBLEMS

2. National Scan, Inc., sells radio frequency inventory tags. Monthly sales for a seven-month period were as follows:

Month	Sales (000 units)
Feb.	19
Mar.	18
Apr.	15
May	20
Jun.	18
Jul.	22
Aug.	20

a. Plot the monthly data on a sheet of graph paper.

b. Forecast September sales volume using each of the following:

(1) The naïve approach
(2) A five-month moving average
(3) A weighted average using .60 for August, .30 for July, and .10 for June
(4) Exponential smoothing with a smoothing constant equal to .20, assuming a March forecast of 19(000)
(5) A linear trend equation

c. Which method seems least appropriate? Why? (*Hint:* Refer to your plot from part a.)

d. What does use of the term *sales* rather than *demand* presume?

4. An electrical contractor's records during the last five weeks indicate the number of job requests:

Week	1	2	3	4	5
Requests	20	22	18	21	22

Predict the number of requests for week 6 using each of these methods:

a. Naive.
b. A four-period moving average.
c. Exponential smoothing with $\alpha = .30$; use 20 for week 2 forecast.

5. A cosmetic manufacturer's marketing department has developed a linear trend equation that can be used to predict annual sales of its Hand & Foot Cream.

$F_t = 80 + 15t$

where

F_t = Annual sales (000 bottles)

t is in years

a. Are annual sales increasing or decreasing? By how much?
b. Predict annual sales for year 6 using the equation.

8. Air Travel on Mountain Airlines for the past 18 weeks was:

Week	Passengers
1	405
2	410
3	420
4	415
5	412
6	420
7	424
8	433
9	438
10	440
11	446
12	451
13	455
14	464
15	466
16	474
17	476
18	482

a. Explain why an averaging technique would not be appropriate for forecasting.
b. Use an appropriate technique to develop a forecast for the expected number of passengers for the next three weeks.

21. Two different forecasting techniques (F_1 and F_2) were used to forecast demand for cases of bottled water. Actual demand and the two sets of forecasts are as follows:

		PREDICTED DEMAND	
Period	**Demand**	**F_1**	**F_2**
1	68	66	66
2	75	68	68
3	70	72	70
4	74	71	72
5	69	72	74
6	72	70	76
7	80	71	78
8	78	74	80

a. Compute MAD for each set of forecasts. Given your results, which forecast appears to be more accurate? Explain.
b. Compute the MSE for each set of forecasts. Given your results, which forecast appears to be more accurate?
c. In practice, *either* MAD *or* MSE would be employed to compute forecast errors. What factors might lead a manager to choose one rather than the other?
d. Compute MAPE for each data set. Which forecast appears to be more accurate?